

Convergence Rate of a Multigrid Method with Gauss-Seidel Relaxation for the Poisson Equation

1. Setup

Poisson equation with Dirichlet boundary conditions

$$\begin{aligned} -\Delta u &= f & \text{in } \Omega \\ u &= 0 & \text{on } \partial\Omega \end{aligned} \quad (1)$$

2. Grids

$$\begin{aligned} \Omega_h &: \text{Dreiecke mit Kantenlängen } h \text{ und } \sqrt{2}h \\ \Omega_H &: \text{Dreiecke mit Kantenlängen } \sqrt{2}h \text{ und } 2h \\ \Omega_{2h} &: \text{Dreiecke mit Kantenlängen } 2h \text{ und } 2\sqrt{2}h \end{aligned} \quad (2)$$

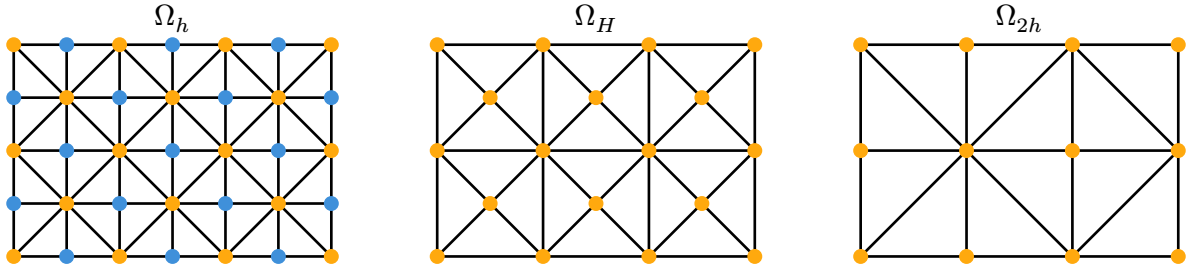


Figure 1: “The grid is divided into fine points (blue) and coarse points (yellow)”

3. Decomposition

$$\begin{aligned} S_h &: \text{finite Elemente Raum auf } \Omega_h \\ S_H &: \text{finite Elemente Raum auf } \Omega_H \\ T_h &:= \{w \in S_h \mid w(p) = 0 \quad \forall p \in \Omega_H\} \end{aligned} \quad (3)$$

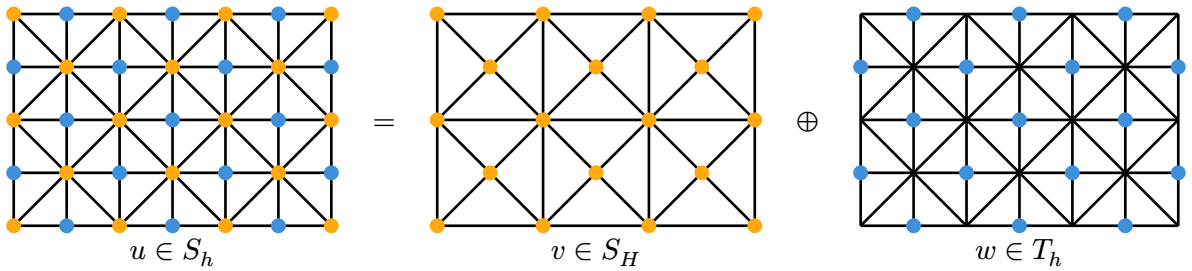


Figure 2: The space S_h of the fine grid can be expressed as a sum of the space S_H on the coarse grid (middle) and the space T_h (right).

$$S_h = S_H \oplus T_h \quad u = v + w \quad (4)$$

The basis functions of T_h form pyramid-like patterns. As a result, all functions in T_h are zero on the diagonal lines through the coarse grid points.

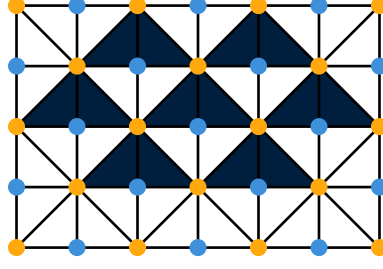


Figure 3: Each shaded are corresponds to one of the basis functions of T_h . The supports of the respective basis functions are disjoint (they do not overlap each other).

4. Norms

For $u, v \in S_h$, we define the bilinear form

$$a(u, v) := \int_{\Omega} (u_{\xi} v_{\xi} + u_{\eta} v_{\eta}) \, d\xi \, d\eta. \quad (5)$$

Furthermore, the energy norm is given by

$$\|v\| := \sqrt{a(v, v)}. \quad (6)$$

Therefore,

$$\begin{aligned} \|v\|^2 = a(v, v) &= \int_{\Omega} v_{\xi}^2 + v_{\eta}^2 \\ &= \sum_{\nu} \int_{\nu} v_{\xi}^2 + v_{\eta}^2 \\ &= \sum_{\nu} \left(v_{\xi}^2|_{\nu} + v_{\eta}^2|_{\nu} \right) \underbrace{\frac{1}{2} h^2}_{\text{area}(\nu)} \\ &= \sum_{\nu} \left(\frac{(v_{\text{east}} - v_{\text{west}})^2}{h^2} + \frac{(v_{\text{north}} - v_{\text{south}})^2}{h^2} \right) \frac{1}{2} h^2 \\ &= \sum_{\nu} \frac{1}{2} (v_{\text{east}} - v_{\text{west}})^2 + \sum_{\nu} \frac{1}{2} (v_{\text{north}} - v_{\text{south}})^2 \\ &= \sum_{\substack{i, j \in \Omega_h \\ d(i, j) = h}} (v_i - v_j)^2. \end{aligned} \quad (7)$$

The last step is justified because each vertical and horizontal border is adjacent to two different triangles (accounting for the factor 2). Note that each pair (i, j) is only used once, meaning that if we include pair (i, j) in the sum ($i \neq j$), we do not include (j, i) as well.

Similarly, we define a seminorm (i.e., something that looks like a norm but that could evaluate to zero, $|x| = 0$, even though $x \neq 0$),

$$|v|^2 := \sum_{\substack{i, j \in \Omega_H \\ d(i, j) = 2h}} \frac{1}{2} (v_i - v_j)^2. \quad (8)$$

Note that for both norms the sum is only taken over all vertical and horizontal edges of the triangulation (i.e., pairs (i, j) with Euclidean distance h), not over the diagonal edges.

For any $p_i, p_j \in \Omega_H$ with $d(i, j) = 2h$, the midpoint p_k between them lies in Ω_h and satisfies $d(i, k) = d(k, j) = h$. Since

$$(u_i - u_j)^2 = ((u_i - u_k) + (u_k - u_j))^2 \leq 2(u_i - u_k)^2 + 2(u_k - u_j)^2 \quad (9)$$

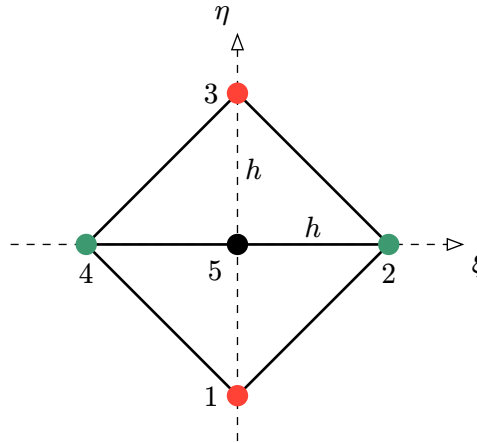
(this follows from $((u_i - u_k) - (u_k - u_j))^2 \geq 0$), summing over all such pairs gives

$$|u|^2 = \sum_{\substack{i, j \in \Omega_H \\ d(i, j) = 2h}} \frac{1}{2} (u_i - u_j)^2 \leq \sum_{\substack{i, j \in \Omega_H \\ d(i, j) = 2h}} (u_i - u_k)^2 + (u_k - u_j)^2. \quad (10)$$

Every point $p_k \in \Omega_h \setminus \Omega_H$ has exactly two $2h$ -edges of Ω_H passing through it, contributing exactly its four incident h -edges of Ω_h . Hence, summing over all pairs $(i, j) \in \Omega_H$, every h -edge (i, k) with $p_k \in \Omega_h \setminus \Omega_H$ appears exactly once on the right-hand side, so it equals $\|u\|^2$. Therefore,

$$|u| \leq \|u\| \quad \forall u \in S_h. \quad (11)$$

5. Estimating $a(v, w)$



Here, $I = \triangle 124$ and $II = \triangle 423$.

$$a(v, w)_I = \int_I v_\xi w_\xi + v_\eta w_\eta = \int_I v_\xi w_\xi + \int_I v_\eta w_\eta \quad (12)$$

$$\int_I v_\xi w_\xi = \int_{I_{\text{links}}} v_\xi w_\xi + \int_{I_{\text{rechts}}} v_\xi w_\xi = 0 \quad (13)$$

da $w_\xi|_{I_{\text{links}}} = -w_\xi|_{I_{\text{rechts}}}$ und $v_\xi = \text{const.}$

$$\begin{aligned}
\int_I v_\eta w_\eta &= \int_I \frac{1}{2} v_\eta^2 - \frac{1}{2} v_\eta^2 + \frac{1}{2} w_\eta^2 - \frac{1}{2} w_\eta^2 + v_\eta w_\eta \\
&= \int_I \frac{1}{2} (-v_\eta^2 + 2v_\eta w_\eta - w_\eta^2) + \frac{1}{2} v_\eta^2 + \frac{1}{2} w_\eta^2 \\
&= \int_I \frac{1}{2} v_\eta^2 + \frac{1}{2} w_\eta^2 - \frac{1}{2} (v_\eta^2 - 2v_\eta w_\eta + w_\eta^2) \\
&= \frac{1}{2} \int_I v_\eta^2 + \frac{1}{2} \int_I w_\eta^2 - \frac{1}{2} \int_I (v_\eta + w_\eta)^2 \\
&= \frac{1}{2} \int_I v_\eta^2 + \frac{1}{2} \int_I v_\xi^2 - \frac{1}{2} \int_I v_\xi^2 + \frac{1}{4} \int_I (w_\eta^2 + w_\xi^2) - \frac{1}{2} \int_I (v_\eta + w_\eta)^2 \\
&= \underbrace{\frac{1}{2} \int_I (v_\eta^2 + v_\xi^2)}_{A_I} + \underbrace{\frac{1}{4} \int_I (w_\eta^2 + w_\xi^2)}_{B_I} - \underbrace{\frac{1}{2} \int_I v_\xi^2}_{C_I} - \underbrace{\frac{1}{2} \int_I (v_\eta + w_\eta)^2}_{D_I} \\
&= A_I + B_I - C_I - D_I
\end{aligned} \tag{14}$$

When summing up the term A for all triangles in the triangulation, we simply get

$$\frac{1}{2} \int_\Omega v_\eta^2 + v_\xi^2 = \frac{1}{2} a(v, v) = \frac{1}{2} \|v\|^2. \tag{15}$$

Similarly, for B :

$$\frac{1}{4} \int_\Omega w_\eta^2 + w_\xi^2 = \frac{1}{4} a(w, w) = \frac{1}{4} \|w\|^2. \tag{16}$$

Since v is linear in both $I = \triangle 124$ and in $\Pi = \triangle 423$, v_ξ is a constant in both triangles. However, because of the shared boundary $\overline{42}$, we know that $v_\xi|_I = v_\xi|_\Pi = \frac{v_2 - v_4}{2h}$.

For C we find

$$\sum_\nu C_\nu = \sum_{\substack{i, j \text{ green} \\ d(i, j) = 2h}} \frac{1}{4} (v_i - v_j)^2 \tag{17}$$

We can also show that

$$\sum_\nu D_\nu \geq \sum_{\substack{i, j \text{ red} \\ d(i, j) = 2h}} \frac{1}{4} (v_i - v_j)^2 \tag{18}$$

Together with C , we get

$$\begin{aligned}
\sum_{\nu} C_{\nu} + D_{\nu} &\geq \sum_{\substack{i,j \text{ green} \\ d(i,j)=2h}} \frac{1}{4} (v_i - v_j)^2 + \sum_{\substack{i,j \text{ red} \\ d(i,j)=2h}} \frac{1}{4} (v_i - v_j)^2 \\
&= \sum_{\substack{i,j \in \Omega_H \\ d(i,j)=2h}} \frac{1}{4} (v_i - v_j)^2 \\
&= \frac{1}{2} \sum_{\substack{i,j \in \Omega_H \\ d(i,j)=2h}} \frac{1}{2} (v_i - v_j)^2 \\
&= \frac{1}{2} |v|^2
\end{aligned} \tag{19}$$

according to the definition of the seminorm $|\cdot|$.

When summing up the terms for A (Equation 15), B (Equation 16), C and D (Equation 19), we find that

$$a(v, w) \leq \underbrace{\frac{1}{2} \|v\|^2}_A + \underbrace{\frac{1}{4} \|w\|^2}_B - \underbrace{\frac{1}{2} |v|^2}_{C+D} = \frac{1}{2} (\|v\|^2 - |v|^2) + \frac{1}{4} \|w\|^2. \tag{20}$$

Notice that we have an inequality ($\leq$) here because of the bound on $\sum_{\nu} D_{\nu}$.

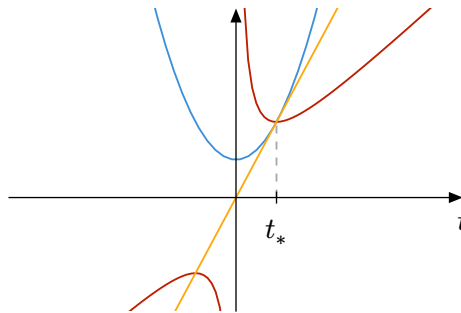
Interestingly, this upper bound is quadratic in $\|w\|$ but at the same time, bilinear form $a(v, w)$ is linear in w since $a(v, tw) = t a(v, w)$ for $t \in \mathbb{R}$ (homogeneity). If we set $w =: t\hat{w}$ such that $\|\hat{w}\| = 1$ and $t > 0$, we get

$$\begin{aligned}
a(v, t\hat{w}) &= t a(v, \hat{w}) \leq \frac{1}{2} (\|v\|^2 - |v|^2) + \frac{1}{4} t^2 \quad | : t \\
a(v, \hat{w}) &\leq \frac{\|v\|^2 - |v|^2}{2t} + \frac{1}{4} t.
\end{aligned} \tag{21}$$

We substitute $\mu := \|v\|^2 - |v|^2$ for brevity. This function has the following minimum

$$\begin{aligned}
\frac{\partial}{\partial t} \left(\frac{\mu}{2t} + \frac{1}{4} t \right) &= -\frac{\mu}{2t^2} + \frac{1}{4} \stackrel{!}{=} 0 \\
t_* &= \sqrt{2\mu}.
\end{aligned} \tag{22}$$

This is the value where the tangent $f(t) = Ct$ (yellow) touches the quadratic function $\frac{1}{2}\mu + \frac{1}{4}t^2$ (blue). The derivative is shown in red.



Accordingly,

$$\begin{aligned}
a(v, \hat{w}) &\leq \frac{\mu}{2t_*} + \frac{1}{4}t_* \\
&= \frac{\mu}{2\sqrt{2\mu}} + \frac{1}{4}\sqrt{2\mu} \\
&= \frac{1}{\sqrt{8}}\sqrt{\mu} + \frac{\sqrt{2}}{\sqrt{16}}\sqrt{\mu} \\
&= \sqrt{\frac{1}{2}\mu}.
\end{aligned} \tag{23}$$

Now we multiply on both sides by $t = \|w\| > 0$ and we get

$$t a(v, \hat{w}) = a(v, t \hat{w}) = a(v, w) \leq \sqrt{\frac{1}{2}\mu} \|w\|. \tag{24}$$

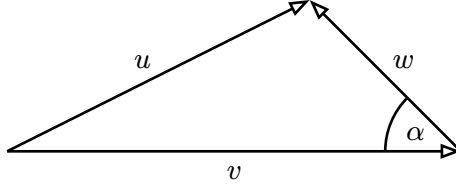
Because of homogeneity, when we negate w , we get

$$-a(v, w) = a(v, -w) \leq \sqrt{\frac{1}{2}\mu} \|-w\| = \sqrt{\frac{1}{2}\mu} \|w\|. \tag{25}$$

Therefore, we can replace $a(v, w)$ by its absolute value. Finally, this yields

$$|a(v, w)| \leq \sqrt{\frac{1}{2} (\|v\|^2 - |v|^2)} \|w\| = \sqrt{\frac{1}{2} \left(1 - \frac{|v|^2}{\|v\|^2}\right)} \|v\| \|w\|. \tag{26}$$

Geometric Interpretation



We decompose u into the sum $v + w$. These vectors are part of the same Hilbert space with inner product $a(\cdot, \cdot)$. We define the angle α as

$$\cos(\alpha) = \frac{a(v, w)}{\|v\| \|w\|}. \tag{27}$$

Therefore, lemma 3.1 can be written in the following way,

$$|\cos(\alpha)| = \frac{|a(v, w)|}{\|v\| \|w\|} \leq \sqrt{\frac{1}{2} \left(1 - \left(\frac{|v|}{\|v\|}\right)^2\right)} \tag{28}$$

or, alternatively,

$$\cos^2(\alpha) \leq \frac{1}{2}(1 - \lambda^2) \tag{29}$$

where $\lambda := \frac{|v|}{\|v\|}$. Due to the identity $\sin^2 \alpha + \cos^2 \alpha = 1$, we can rearrange this inequality,

$$\cos^2 \alpha = 1 - \sin^2 \alpha \leq \frac{1}{2}(1 - \lambda^2) \implies \sin^2 \alpha \geq \frac{1}{2}(1 + \lambda^2). \tag{30}$$

Furthermore, we observe that for a fixed v and a fixed α , u is the shortest when $u \perp w$. In this case, $\sin(\alpha) = \frac{\|u\|}{\|v\|}$. Generally, $\|u\| \geq \|v\| \sin(\alpha)$. Since u and v have the same values on the green and red points, we know that $|u| = |v|$.

$$|u| = |v| = \lambda \|v\| \leq \frac{\lambda}{\sin(\alpha)} \|u\| \leq \frac{\lambda}{\sqrt{\frac{1}{2}(1 + \lambda^2)}} \|u\| \quad (31)$$

Finally, we express this as

$$\frac{|u|}{\|u\|} \leq \frac{\lambda}{\sqrt{\frac{1}{2}(1 + \lambda^2)}} = \sqrt{\frac{\lambda^2}{\frac{1}{2}(1 + \lambda^2)}} = \sqrt{\frac{2\lambda^2}{1 + \lambda^2}} =: \sqrt{\rho} \quad (32)$$

6. Gauß-Seidel Relaxation

We use a finite element discretization. Interestingly, the stencil for the black all points is the same, wheter in Ω_h or in Ω_H :

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix} \quad (33)$$

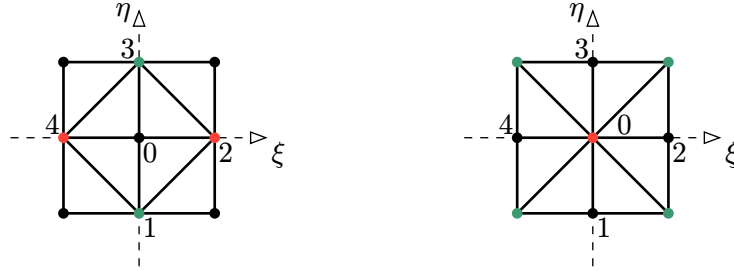


Figure 7: Neighborhood of a fine grid node in $\Omega_h \setminus \Omega_H$ (left) and of a coarse node Ω_H (right).

The stencil results in the following Gauss-Seidel relaxation for a single row:

$$u_i = \frac{1}{4} \sum_{j=1}^4 u_j + b_i \quad (34)$$

where $b_i = \int_{\Omega} f \varphi_i$ is the right-hand side. In the following section, we will be concerned with the error of our approximation for u . Therefore, we assume that $b = 0$.

We split the Gauss-Seidel relaxation into two steps ($\triangleq$ red-black Gauss-Seidel). First, we will smooth the fine points ($\in \Omega_h \setminus \Omega_H$), then the coarse points ($\in \Omega_H$). Formally,

$$\begin{aligned} (G_h^I u)_i &= \begin{cases} u_i & p_i \in \Omega_H \\ \frac{1}{4} \sum_{j \in N(i)} u_j & p_i \in \Omega_h \setminus \Omega_H \end{cases} \\ (G_h^{II} u)_i &= \begin{cases} \frac{1}{4} \sum_{j \in N(i)} u_j & p_i \in \Omega_H \\ u_i & p_i \in \Omega_h \setminus \Omega_H \end{cases} \end{aligned} \quad (35)$$

Here, $N(i)$ is the set of neighboring points with distance h to p_i .

Lemma 4.1

Assume that Ω is convex. We want to show: if $u = G_h^I u$ then

$$\|G_h^{II} u\| \leq |u|. \quad (36)$$

half a Gauss-Seidel step decreases the energy norm beyond the seminorm - this means it is reduced by at least the factor $\sqrt{\rho}$.

Proof. For convenience we first ignore the boundary $\partial\Omega$; we comment on the necessary modifications at the end of the proof.

Let $\bar{u} := G_h^\Pi u$, and fix a point $p_0 \in \Omega_H$ with its four neighbors 1, 2, 3, 4 at distance h , which necessarily lie in $\Omega_h \setminus \Omega_H$ (cf. Figure 7, right).

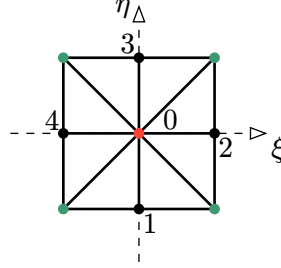


Figure 8: Point $p_0 \in \Omega_H$ and its four neighbors $1, 2, 3, 4 \in \Omega_h \setminus \Omega_H$ at distance h

For arbitrary real numbers z_1, z_2, z_3, z_4 one has the algebraic identity

$$2 \sum_{j=1}^4 z_j^2 = (z_1 - z_2)^2 + (z_2 - z_3)^2 + (z_3 - z_4)^2 + (z_4 - z_1)^2 + \frac{1}{2}(z_1 + z_2 + z_3 + z_4)^2 - \frac{1}{2}(z_1 - z_2 + z_3 - z_4)^2. \quad (37)$$

Since $\bar{u} = G_h^\Pi u$, the value at p_0 is by definition the mean of the (unaffected) values at its neighbors,

$$\bar{u}_0 = \frac{1}{4} \sum_{i=1}^4 \bar{u}_i = \frac{1}{4} \sum_{i=1}^4 u_i, \quad (38)$$

because G_h^Π leaves the values at $\Omega_h \setminus \Omega_H$ unchanged. Put $z_i := \bar{u}_i - \bar{u}_0 = u_i - \bar{u}_0$. Then

$$\sum_{j=1}^4 z_j = \sum_{j=1}^4 u_j - 4\bar{u}_0 = 0, \quad (39)$$

so the first-order term $\frac{1}{2}(z_1 + z_2 + z_3 + z_4)^2$ in Equation 37 vanishes, and dropping the remaining non-positive term $-\frac{1}{2}(z_1 - z_2 + z_3 - z_4)^2$ only weakens the identity into an inequality. We obtain

$$\sum_{j=1}^4 (\bar{u}_j - \bar{u}_0)^2 \leq \frac{1}{2} [(u_1 - u_2)^2 + (u_2 - u_3)^2 + (u_3 - u_4)^2 + (u_4 - u_1)^2]. \quad (40)$$

Recall from the definition of the energy norm (2.3) that, splitting the sum according to which grid the points belong to,

$$\|\bar{u}\|^2 = \sum_{i \in \Omega_H} \sum_{\substack{j \in \Omega_h \setminus \Omega_H \\ d(i,j)=h}} (\bar{u}_i - \bar{u}_j)^2. \quad (41)$$

For fixed $i = p_0$, the inner sum is exactly the left-hand side of Equation 40. Summing the bound Equation 40 over all $p_0 \in \Omega_H$ therefore gives

$$\|G_h^\Pi u\|^2 \leq \sum_{\substack{k,l \in \Omega_h \setminus \Omega_H \\ d(k,l)=H}} (u_k - u_l)^2, \quad (42)$$

where $H = \sqrt{2}h$ is the distance between two opposite corners of the diamond in Figure 8.

Every pair $(k, l) \in \Omega_h \setminus \Omega_H$ with $d(k, l) = H$ occurs in the diamonds of **two** neighboring points $p_0 \in \Omega_H$. The factor $\frac{1}{2}$ on the right-hand side of Equation 40 exactly compensates for this double counting, so no extra factor appears in Equation 42.

This is the only place where the hypothesis $u = G_h^I u$ enters. Since $u = G_h^I u$, every point $k \in \Omega_h \setminus \Omega_H$ already satisfies the relaxation equation

$$u_k = \frac{1}{4} \sum_{j \in N(k)} u_j, \quad (43)$$

i.e. u_k is the mean of its four neighbors, which lie in Ω_H .

Consider a pair $k, l \in \Omega_h \setminus \Omega_H$ with $d(k, l) = H$, and let $1, 3, 5, 7 \in \Omega_H$ be enumerated so that 1, 5 are the neighbors shared in the relevant direction (cf. Figure 4 of Braess' paper). Using the mean-value property at k and l ,

$$16(u_k - u_l)^2 = (u_1 + u_5 - u_3 - u_7)^2 \leq 2(u_1 - u_3)^2 + 2(u_5 - u_7)^2, \quad (44)$$

where the last step is the elementary inequality $(a + b)^2 \leq 2a^2 + 2b^2$ applied to $a = u_1 - u_3$, $b = u_5 - u_7$.

Summing this over all pairs (k, l) from Equation 42 – each pair $(u_1 - u_3)^2$ now reappears for two different pairs (k, l) , which compensates the factor $\frac{1}{4}$ – we obtain

$$\|G_h^{\Pi} u\|^2 \leq \frac{1}{4} \sum_{\substack{m, n \in \Omega_H \\ d(m, n) = 2H}} (u_m - u_n)^2. \quad (45)$$

The points m, n are again in Ω_H , but now at distance $2H = 2\sqrt{2}h$: these are exactly the two diagonals of a square of side length $2h$ with corners in Ω_H .

It remains to bound such diagonals by sides of length $2h$. Let $1, 2, 3, 4 \in \Omega_H$ be the corners of such a square, in cyclic order, so that $(u_1 - u_3)^2$ and $(u_2 - u_4)^2$ are the two diagonal terms appearing in Equation 45. A direct expansion gives the identity

$$(u_1 - u_3)^2 + (u_2 - u_4)^2 = (u_1 - u_2)^2 + (u_2 - u_3)^2 + (u_3 - u_4)^2 + (u_4 - u_1)^2 - (u_1 - u_2 + u_3 - u_4)^2, \quad (46)$$

and dropping the (non-positive) last term yields

$$(u_1 - u_3)^2 + (u_2 - u_4)^2 \leq (u_1 - u_2)^2 + (u_2 - u_3)^2 + (u_3 - u_4)^2 + (u_4 - u_1)^2, \quad (47)$$

i.e. the sum of the two diagonals is bounded by the sum of the four sides of the square (each of length $2h$).

Every side of length $2h$ is shared by exactly two neighboring squares, so summing this inequality over all squares with corners in Ω_H and combining with Equation 45 gives

$$\|G_h^{\Pi} u\|^2 \leq \frac{1}{2} \sum_{\substack{n, m \in \Omega_H \\ d(n, m) = 2h}} (u_n - u_m)^2 = |u|^2, \quad (48)$$

which is exactly the definition of the seminorm $|\cdot|$ from (2.4). This proves

$$\|G_h^{\text{II}} u\| \leq |u|. \quad (49)$$

Near $\partial\Omega$, some of the neighbors 1, 2, 3, 4 (or 1, 3, 5, 7) used above may not exist inside Ω ; by the homogeneous Dirichlet condition, the corresponding values are simply replaced by 0. The same telescoping arguments go through verbatim with these zero values inserted — only the index bookkeeping becomes more tedious, while every inequality used above (Equation 37, the mean-value property, the diagonal identity) remains valid for boundary configurations as well, consistent with how the doubled weight in (2.5) was introduced for the seminorm. ■

As a consequence of Lemma 4.1 we have the following corollary.

Corollary 4.2. If Ω is convex, then $\|G_h^{\text{II}} \cdot G_h^{\text{I}} u\| \leq |u|$ for every $u \in S_h$ — i.e. the hypothesis $u = G_h^{\text{I}} u$ in Lemma 4.1 can be dropped, at the price of applying G_h^{I} first.

Proof. Put $u' := G_h^{\text{I}} u$. Since G_h^{I} is idempotent ($G_h^{\text{I}} G_h^{\text{I}} = G_h^{\text{I}}$), we have $u' = G_h^{\text{I}} u'$, so Lemma 4.1 applies to u' :

$$\|G_h^{\text{II}} u'\| \leq |u'|. \quad (50)$$

Now G_h^{I} only changes the w -part of the decomposition $u = v + w$ (it leaves the values at Ω_H , i.e. the v -part, untouched), and the seminorm $|\cdot|$ depends only on the v -part by (3.3). Hence

$$|u'| = |G_h^{\text{I}} u| = |u|, \quad (51)$$

and combining the two displays gives $\|G_h^{\text{II}} G_h^{\text{I}} u\| \leq |u|$ for every $u \in S_h$. ■

We note that the constant 1 in (4.2) is best possible, since the convergence rate of the (pure) Gauss-Seidel iteration is known to be close to 1, with $\|G_h^{\text{II}} \cdot G_h^{\text{I}}\| = 1 - O(h^2)$.

7. Alternate Method

Proof that $G^{\text{I}} = I - P_V$ where P_V is an orthogonal projection and $V = T_h$. The first Gauß-Seidel half-step is associated with the splitting $S_h = S_H \oplus T_h$ where $T_h = \{w \in S_h \mid w(p_i) = 0 \text{ for } p_i \in \Omega_H\}$. All functions in $u + T_h$ have the same values as u at the white nodes (and possibly different values at the black nodes).

Don't call it w since $w \in T_h$ and $G^{\text{I}} u \in T_h^\perp$.

$w := G^{\text{I}} u$ minimizes $J(v)$ over $u + T_h$. This is because of the following reason:

$$J(v) = a(v, v) = \|v\|^2 = \sum_{\substack{i,j \\ d(i,j)=h}} (v_i - v_j)^2. \quad (52)$$

Now, if we look at the support of a single basis function of T_h (see Figure 7 left), we find that the relevant terms of the sum are

$$(v_1 - v_0)^2 + (v_2 - v_0)^2 + (v_3 - v_0)^2 + (v_4 - v_0)^2 = \text{function}(v_0), \quad (53)$$

and the minimum of this function is at $v_0 = \frac{1}{4}(v_1 + v_2 + v_3 + v_4)$, which is exactly what G^{I} computes for v_0 .

Since w minimizes $J(v)$ over $u + T_h$, going in any direction $t \in T_h$ must increase the energy: $J(w + t) \geq J(w) \forall t \in T_h$. This can also be written as

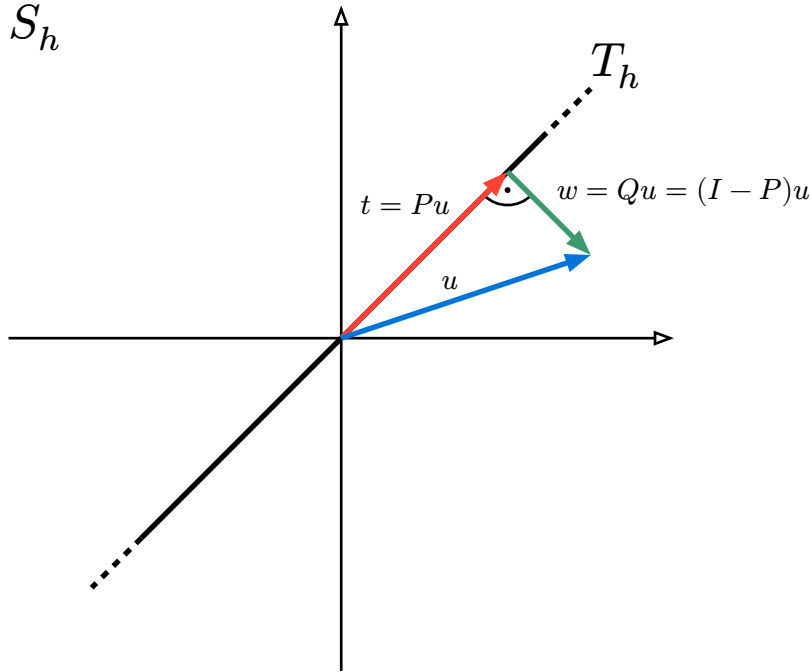
$$\frac{d}{d\varepsilon} J(w + \varepsilon t)|_{\varepsilon=0} \stackrel{!}{=} 0 \quad \forall t \in T_h. \quad (54)$$

Now we use the definition of $J(v)$ and $a(u, v)$ (we assume that the right-hand side $f = 0$).

no need to substitute the definition of a since we know it is linear and symmetric

$$\begin{aligned}
\frac{d}{d\varepsilon} J(w + \varepsilon t)|_{\varepsilon=0} &= \frac{d}{d\varepsilon} a(w + \varepsilon t, w + \varepsilon t)|_{\varepsilon=0} \\
&= \frac{d}{d\varepsilon} \int_{\Omega} \left(\frac{\partial(w + \varepsilon t)}{\partial \xi} \right)^2 + \left(\frac{\partial(w + \varepsilon t)}{\partial \eta} \right)^2 \Big|_{\varepsilon=0} \\
&= \frac{d}{d\varepsilon} \int_{\Omega} (w_{\xi} + \varepsilon t_{\xi})^2 + (w_{\eta} + \varepsilon t_{\eta})^2 \Big|_{\varepsilon=0} \\
&= \frac{d}{d\varepsilon} \int_{\Omega} (w_{\xi}^2 + 2\varepsilon w_{\xi} t_{\xi} + \varepsilon^2 t_{\xi}^2) + (w_{\eta}^2 + 2\varepsilon w_{\eta} t_{\eta} + \varepsilon^2 t_{\eta}^2) \Big|_{\varepsilon=0} \\
&= \frac{d}{d\varepsilon} \left[\int_{\Omega} (w_{\xi}^2 + w_{\eta}^2) + 2\varepsilon \int_{\Omega} (w_{\xi} t_{\xi} + w_{\eta} t_{\eta}) + \varepsilon^2 \int_{\Omega} (t_{\xi}^2 + t_{\eta}^2) \right]_{\varepsilon=0} \\
&= \frac{d}{d\varepsilon} [a(w, w) + 2\varepsilon a(w, t) + \varepsilon^2 a(t, t)]_{\varepsilon=0} \\
&= [2a(w, t) + 2\varepsilon a(t, t)]_{\varepsilon=0} \\
&= 2a(w, t) \\
&= 2(w, t) \stackrel{!}{=} 0 \quad \forall t \in T_h
\end{aligned} \quad (55)$$

This inner product being 0 means that w is perpendicular to T_h , or, $w \in T_h^{\perp}$.



Since G^I only changes the values at the black nodes, we can find a $t \in T_h$ such that $u = w + t$. According to the orthogonal projection theorem, for a given vector u , vectors w and t are unique. Since $w \in T_h^{\perp}$ and $t \in T_h$, we can view t as the orthogonal projection of $u \in S_h$ onto the subspace T_h .

This is written as $Pu = t$ where P is the projection operator onto T_h . Similarly, we have $Q := I - P$ (where I is the identity operator), such that

$$Qu = (I - P)u = Iu - Pu = u - t = w = G^I u. \quad (56)$$

Therefore, $Q = G^I$.

formal falsch, weil G^I auf Vektoren arbeitet und Q auf Funktionen (?)

Now we want to show that $Q = G^I$ is self-adjoint, i.e.,

$$(Qu, v) = (u, Qv) \quad \forall u, v \in S_h. \quad (57)$$

then be directly shown for Q as a projection onto $T_h^\perp$

Let $u = u_1 + u_2$ where $u_1 \in T_h$ and $u_2 \in T_h^\perp$. Similarly, $v = v_1 + v_2$ where $v_1 \in T_h$ and $v_2 \in T_h^\perp$.

$$\begin{aligned} (Pu, v) &= (P(u_1 + u_2), v) = (Pu_1 + Pu_2, v) = (u_1, v) = (u_1, v_1 + v_2) \\ &= (u_1, v_1) + \underbrace{(u_1, v_2)}_{=0 \text{ since } u_1 \perp v_2} = (u_1, v_1) \end{aligned} \quad (58)$$

$$\begin{aligned} (u, Pv) &= (u, P(v_1 + v_2)) = (u, Pv_1 + Pv_2) = (u, v_1) = (u_1 + u_2, v_1) \\ &= (u_1, v_1) + \underbrace{(u_2, v_1)}_{=0 \text{ since } u_2 \perp v_1} = (u_1, v_1) \end{aligned} \quad (59)$$

Therefore, $(Pu, v) = (u, Pv)$. The same is true for Q :

$$\begin{aligned} (Qu, v) &= ((I - P)u, v) = (u - Pu, v) = (u, v) - (Pu, v) = (u, v) - (u, Pv) \\ &= (u, v - Pv) = (u, (I - P)v) = (u, Qv). \end{aligned} \quad (60)$$

The same can be shown for G^{II} , only that here the relevant subspace is $\hat{T}_h := \{w \in S_h \mid w(p_i) = 0 \text{ for } p_i \in \Omega_h \setminus \Omega_H\}$. Note that both G^I and G^{II} are idempotent. This means that if we apply the same orthogonal projection twice, the result will be the same as when we only apply it once ($G^I G^I = G^I$, or, $G^I G^I u = G^I u$). Assuming that $G^{\text{II}} u = u$,

$$\begin{aligned} \|G^I u\|^2 &= (G^I u, G^I u) = (u, G^I G^I u) = (u, G^I u) \\ &= (G^{\text{II}} u, G^I u) = (u, G^{\text{II}} G^I u) \leq \|u\| \cdot \|G^{\text{II}} G^I u\|. \end{aligned} \quad (61)$$

Hence,

$$\frac{\|G^I u\|}{\|u\|} \leq \frac{\|G^{\text{II}} G^I u\|}{\|G^I u\|}. \quad (62)$$

The same can be shown the other way around. Assuming $G^I u = u$,

$$\frac{\|G^{\text{II}} u\|}{\|u\|} \leq \frac{\|G^I G^{\text{II}} u\|}{\|G^{\text{II}} u\|}. \quad (63)$$

In other words, the norm reduction becomes less effective with each half-step. (Of course, the smaller $\frac{\|G^{I/\text{II}} u\|}{\|u\|}$ the better, since the solution to the homogeneous equation $\Delta u = 0$ with a zero Dirichlet boundary condition is $u = 0$).

8. Multigrid Algorithm

First iteration on level q with $r = 2$ smoothings:

1. $u^{q,0} \leftarrow G_q^I u^{q,0}$ (special case because this is the first iteration)
2. $u^{q,1} \leftarrow G_q^{II} u^{q,0}$
3. $u^{q,2} \leftarrow G_q^I u^{q,1}$ (transition step)
4. approximate v_1 as the error $e \in S_{q-1}$ of the variational problem $J(u^{q,2} + e) \rightarrow \min$
 - on the coarsest level $q = 1$, compute v_1 exactly
 - otherwise, use a multigrid iteration on level $q - 1$, starting with $u^{q-1,0} = 0$
5. $u^{q,3} \leftarrow u^{q,2} + v_1$ (correction)
6. $u^{q,5} \leftarrow G_q^I u^{q,4}$ (mistake in the indices)
7. $u^{q,6} \leftarrow G_q^{II} u^{q,5}$
8. $u^{q,7} \leftarrow G_q^I u^{q,6}$
9. start the next iteration using $u^{q,0} \leftarrow u^{q,6}$ (even though we just computed $u^{q,7}$)

probably wrong: 6. should probably start with G_q^{II}

Next iterations omit step 1. and directly start with $u^{q,1} = G_q^{II} u^{1,0}$.

$$u^0 \xrightarrow{G^I} u^1 \xrightarrow{G^{II}} u^2 \xrightarrow{G^I} u^3 \xrightarrow{\text{coarse grid}} u^4 \xrightarrow{G^I} u^5 \xrightarrow{G^{II}} u^6 \xrightarrow{G^I} u^7 \quad (64)$$

First iteration on level q with $r = 3$ smoothings:

1. $u^{q,0} \leftarrow G_q^{II} u^{q,0}$ (special case because this is the first iteration)
2. $u^{q,1} \leftarrow G_q^I u^{q,0}$
3. $u^{q,2} \leftarrow G_q^{II} u^{q,1}$
4. $u^{q,3} \leftarrow G_q^I u^{q,2}$
5. coarse grid correction
6. $u^{q,4} \leftarrow u^{q,3} + v_1$
7. $u^{q,6} \leftarrow G_q^I u^{q,5}$ (mistake in the indices)
8. $u^{q,7} \leftarrow G_q^{II} u^{q,6}$
9. $u^{q,8} \leftarrow G_q^I u^{q,7}$
10. $u^{q,9} \leftarrow G_q^{II} u^{q,8}$
11. start the next iteration using $u^{q,0} \leftarrow u^{q,8}$

$$u^0 \xrightarrow{G^{II}} u^1 \xrightarrow{G^I} u^2 \xrightarrow{G^{II}} u^3 \xrightarrow{G^I} u^4 \xrightarrow{\text{coarse grid}} u^5 \xrightarrow{G^I} u^6 \xrightarrow{G^{II}} u^7 \xrightarrow{G^I} u^8 \xrightarrow{G^{II}} u^9 \quad (65)$$

9. Actual Proof

If we look at the sequence of Gauss-Seidel relaxations as an alternate method where $Q_V = G_q^I$ and $Q_W = G_q^{II}$,

why can we do that? How is $G_q^I = \text{id} - P_V$?

then we notice that the last half-step is the least effective (since the norm reduction decreases from step to step):

$$\frac{\|u^{\nu+1}\|}{\|u^{\nu+\frac{1}{2}}\|} \geq \frac{\|u^{\nu+\frac{1}{2}}\|}{\|u^\nu\|}. \quad (66)$$

This is independent of whether the last half-step is an application of G_q^I or G_q^{II} (since we can swap Q_V and Q_W).

We also found that

$$\frac{\|G_q^{II}u^r\|}{\|u^r\|} \leq \frac{|u^r|}{\|u^r\|} \leq \sqrt{\rho}. \quad (67)$$

This must be true even for the last application of G_q^{II} in the sequence. All previous half-steps must have a norm reduction of $\sqrt{q}$ or better (*better* in this case means *smaller* than $\sqrt{q}$). Therefore,

$$\frac{\|u^r\|}{\|u^0\|} \leq (\sqrt{q})^r = q^{\frac{r}{2}} \quad (68)$$

If $u' = u^r - u^{q-1}$ then

Shouldn't it be $u' = u^r + u^{q-1}$?

proof

$$\begin{aligned} \|u'\| &\leq \sqrt{\frac{1}{2}(1 - \lambda^2)} \cdot \|u^r\| \\ &\leq \rho^{\frac{r}{2}} \sqrt{\frac{1 - \lambda^2}{2}} \cdot \|u^0\| \\ &= \rho^{\frac{r}{2}} \sqrt{\frac{\frac{1 - \lambda^2}{\lambda^2 + 1}}{2}} \cdot \|u^0\| = \rho^{\frac{r}{2}} \sqrt{\frac{\frac{\lambda^2 + 1 - 2\lambda^2}{\lambda^2 + 1}}{2\lambda^2 + 2 - 2\lambda^2}} \cdot \|u^0\| = \rho^{\frac{r}{2}} \sqrt{\frac{1 - \frac{2\lambda^2}{\lambda^2 + 1}}{2 - \frac{2\lambda^2}{\lambda^2 + 1}}} \cdot \|u^0\| \\ &= \rho^{\frac{r}{2}} \sqrt{\frac{1 - \rho}{2 - \rho}} \cdot \|u^0\| \end{aligned} \quad (69)$$

as a reminder:

$$\rho := \frac{2\lambda^2}{\lambda^2 + 1} \quad \lambda := \frac{|v|}{\|v\|}$$

Now we look at the deviation of v_1 from u^{q-1} . We find that

$$u^{r+1} - u' = (u^r + v_1) - (u^r - u^{q-1}) = v_1 + u^{q-1}. \quad (70)$$

In step 3 of the Multigrid algorithm, we require that

$$\|u^{r+1} - u'\| = \|v_1 + u^{q-1}\| \stackrel{!}{<} \delta \|u^{q-1}\|. \quad (71)$$

define δ first

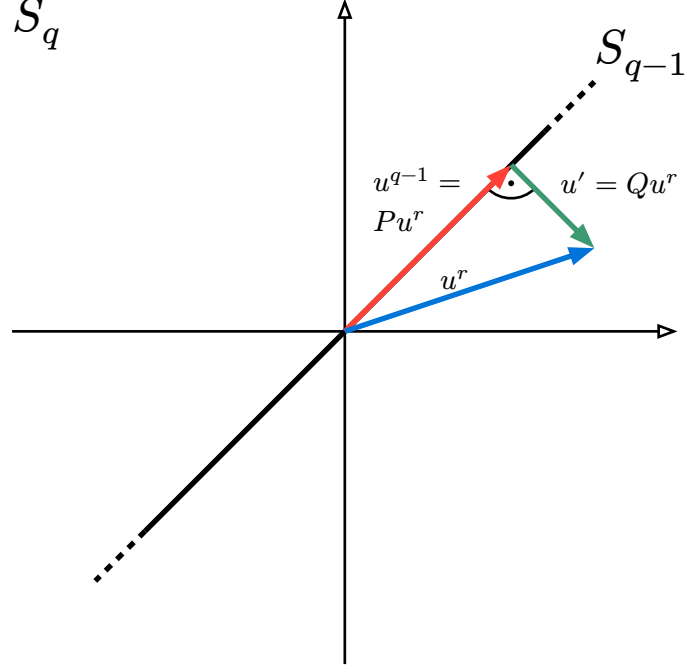
We define

$$v_2 := \frac{1}{\delta} (u^{r+1} - u') = \frac{1}{\delta} (v_1 + u^{q-1}). \quad (72)$$

From Equation 71, we find that

$$\begin{aligned}\|u^{r+1} - u'\| &= \delta \|v_2\| < \delta \|u^{q-1}\| \\ \implies \|v_2\| &< \|u^{q-1}\|\end{aligned}\tag{73}$$

Since u' minimizes $J(v)$ in $u^r + S_{q-1}$, we can show (similarly to Equation 55) that $u' \perp u^{q-1}$. Furthermore, we can apply the operator notation again: $Pu^r = u^{q-1} \in S_{q-1}$ is the projection of u^r onto the subspace S_{q-1} , and $Qu^r = (I - P)u^r = u^r - Pu^r = u' \in S_{q-1}^\perp$.



Since $v_2 \in S_{q-1}$, we find that $v_2 \perp u'$, and therefore we can apply the Pythagorean theorem

$$\|u' + v_2\|^2 = \|u'\|^2 + \|v_2\|^2.\tag{74}$$

Because of Equation 73, we have

$$\|u' + v_2\|^2 = \|u'\|^2 + \|v_2\|^2 \leq \|u'\|^2 + \|u^{q-1}\|^2 = \|u^r\|^2.\tag{75}$$

In both the pre-smoothing and the post-smoothing, we have $r + 1$ alternating factors.

pre-smoothing $\underbrace{G^{r+1} \cdot G^r \cdot \dots \cdot G^2 \cdot G^1}_{=:\tilde{G}}$	coarse grid correction	post-smoothing $\underbrace{G^1 \cdot G^2 \cdot \dots \cdot G^r \cdot G^{r+1}}_{=\tilde{G}^*}$
--	-------------------------------	--

We define

$$G^k := \begin{cases} G^{\text{I}} & k \text{ odd} \\ G^{\text{II}} & k \text{ even} \end{cases}\tag{76}$$

Therefore,

$$u^r = \tilde{G}u^0 \quad \text{and} \quad u^{2r+2} = \tilde{G}^*u^{r+1}.\tag{77}$$

Interestingly, $\tilde{G}^*$ happens to be the adjoint of $\tilde{G}$. This is because of the following reason:

$$\begin{aligned}
(\tilde{G}u, v) &= (G^{r+1}G^rG^{r-1}\dots G^1u, v) = (G^rG^{r-1}\dots G^1u, G^{r+1}v) \\
&= (G^{r-1}\dots G^1u, G^rG^{r+1}v) \\
&= \dots \\
&= (G^1u, G^2\dots G^rG^{r+1}v) \\
&= (u, G^1\dots G^rG^{r+1}v) = (u, \tilde{G}^*v).
\end{aligned} \tag{78}$$

Note that G^k are self-adjoint, as shown in Equation 60.

For an arbitrary $\hat{u} \in S_q$, we find

$$\begin{aligned}
(\hat{u}, u^{2r+2}) &= (\hat{u}, \tilde{G}^*u^{r+1}) \\
&= (\tilde{G}\hat{u}, u^{r+1}) \\
&= (\tilde{G}\hat{u}, u' + \delta v_2) \\
&= (\tilde{G}\hat{u}, u' - \delta u' + \delta u' + \delta v_2) \\
&= (\tilde{G}\hat{u}, [1 - \delta]u' + \delta[u' + v_2]) \\
&= [1 - \delta](\tilde{G}\hat{u}, \underbrace{u'}_{Q \cdot \tilde{G}u^0}) + \delta(\tilde{G}\hat{u}, u' + v_2) \\
&= [1 - \delta](Q\tilde{G}\hat{u}, Q\tilde{G}u^0) + \delta(\tilde{G}\hat{u}, u' + v_2) \\
&\leq [1 - \delta](Q\tilde{G}\hat{u}, Q\tilde{G}u^0) + \delta \|\tilde{G}\hat{u}\| \cdot \|u' + v_2\| \\
&\leq [1 - \delta] \|Q\tilde{G}\hat{u}\| \cdot \|Q\tilde{G}u^0\| + \delta \|\tilde{G}\hat{u}\| \cdot \|u^r\| \\
&= [1 - \delta] \|Q\tilde{G}\hat{u}\| \cdot \|Q\tilde{G}u^0\| + \delta \|\tilde{G}\hat{u}\| \cdot \|\tilde{G}u^0\| \\
&\leq \sqrt{(1 - \delta) \|Q\tilde{G}\hat{u}\|^2 + \delta \|\tilde{G}\hat{u}\|^2} \cdot \sqrt{(1 - \delta) \|Q\tilde{G}u^0\|^2 + \delta \|\tilde{G}u^0\|^2}
\end{aligned} \tag{79}$$

The last transformation step can be explained as follows. If

$$\begin{aligned}
a &:= \sqrt{1 - \delta} \|Q\tilde{G}\hat{u}\|, & c &:= \sqrt{1 - \delta} \|Q\tilde{G}u^0\|, \\
b &:= \sqrt{\delta} \|\tilde{G}\hat{u}\|, & d &:= \sqrt{\delta} \|\tilde{G}u^0\|,
\end{aligned} \tag{80}$$

then

$$\begin{aligned}
[1 - \delta] \|Q\tilde{G}\hat{u}\| \cdot \|Q\tilde{G}u^0\| + \delta \|\tilde{G}\hat{u}\| \cdot \|\tilde{G}u^0\| &= ac + bd = \begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} c \\ d \end{pmatrix} \\
&\leq \left\| \begin{pmatrix} a \\ b \end{pmatrix} \right\| \cdot \left\| \begin{pmatrix} c \\ d \end{pmatrix} \right\| = \sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2} \\
&= \sqrt{(1 - \delta) \|Q\tilde{G}\hat{u}\|^2 + \delta \|\tilde{G}\hat{u}\|^2} \cdot \sqrt{(1 - \delta) \|Q\tilde{G}u^0\|^2 + \delta \|\tilde{G}u^0\|^2}.
\end{aligned} \tag{81}$$

Let's focus on the term $(1 - \delta) \|Q\tilde{G}\hat{u}\|^2 + \delta \|\tilde{G}\hat{u}\|^2$.

as a reminder:

$$\|u^r\| \leq \rho^{\frac{r}{2}} \|u^0\| \quad \text{Equation 68}$$

$$\|u'\| \leq \rho^{\frac{r}{2}} \sqrt{\frac{1-\rho}{2-\rho}} \cdot \|u^0\| \quad \text{Equation 69}$$

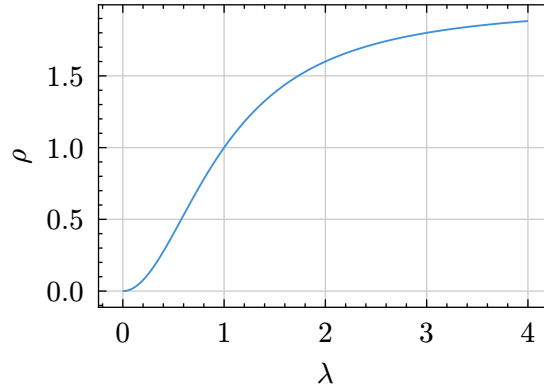
From Equation 68 and Equation 69, we find that

$$\begin{aligned} (1-\delta) \|Q \underbrace{\tilde{G}u^0}_{u^r}\|^2 + \delta \|\tilde{G}u^0\|^2 &= (1-\delta) \|u'\|^2 + \delta \|u^r\|^2 \\ &\leq (1-\delta) \left(\rho^{\frac{r}{2}} \sqrt{\frac{1-\rho}{2-\rho}} \|u^0\| \right)^2 + \delta (\rho^{\frac{r}{2}} \|u^0\|)^2 \\ &= (1-\delta) \rho^r \frac{1-\rho}{2-\rho} \|u^0\|^2 + \delta \rho^r \|u^0\|^2 \\ &= \rho^r \left((1-\delta) \frac{1-\rho}{2-\rho} + \delta \right) \|u^0\|^2 \end{aligned} \quad (82)$$

as a reminder:

$$\rho = \frac{2\lambda^2}{\lambda^2 + 1} \quad \text{and} \quad \lambda = \frac{|v^r|}{\|v^r\|} \quad (83)$$

Note that the starting vector u^0 in Equation 68, Equation 69 and Equation 82 can be set arbitrarily. However, starting from a different u^0 , the resulting $u^r = \tilde{G}u^0$ will be different as well, which leads to a different ρ . Therefore, when we write Equation 82 for an arbitrary $u \in S_q$ as a starting vector, we need to account for all possible values of ρ .



Since $|v^r| \leq \|v^r\|$, we know that $0 \leq \lambda \leq 1$. In this interval, $0 \leq \rho \leq 1$. To preserve the inequality in Equation 82 for an arbitrary starting vector $u \in S_q$, we need to include the upper bound of the term $\rho^r \left((1-\delta) \frac{1-\rho}{2-\rho} + \delta \right)$ on this interval. Hence, we get

$$\begin{aligned} (1-\delta) \|Q\tilde{G}u\|^2 + \delta \|\tilde{G}u\|^2 &\leq \underbrace{\max_{0 \leq \rho \leq 1} \left[\rho^r \left((1-\delta) \frac{1-\rho}{2-\rho} + \delta \right) \right]}_{=: \delta_q} \|u\|^2 \\ &= \delta_q \|u\|^2 \end{aligned} \quad (84)$$

Therefore, we can extend equation Equation 79 as follows:

$$\begin{aligned}
(\hat{u}, u^{2r+2}) &\leq \sqrt{(1-\delta) \|Q\tilde{G}\hat{u}\|^2 + \delta \|\tilde{G}\hat{u}\|^2} \cdot \sqrt{(1-\delta) \|Q\tilde{G}u^0\|^2 + \delta \|\tilde{G}u^0\|^2} \\
&\leq \sqrt{\delta_q \|\hat{u}\|^2} \cdot \sqrt{\delta_q \|u^0\|^2} \\
&= \delta_q \cdot \|\hat{u}\| \cdot \|u^0\| \quad \forall \hat{u} \in S_q.
\end{aligned} \tag{85}$$

This inequality must be true for all $\hat{u} \in S_q$. In order to find an estimate for $\|u^{2r+2}\|$, we use the following trick:

$$\begin{aligned}
(\hat{u}, u^{2r+2}) &\leq \delta_q \|\hat{u}\| \|u^0\| \\
\Rightarrow \frac{(\hat{u}, u^{2r+2})}{\|\hat{u}\|} &\leq \delta_q \|u^0\| \\
\Rightarrow \left(\frac{\hat{u}}{\|\hat{u}\|}, u^{2r+2} \right) &\leq \delta_q \|u^0\|.
\end{aligned} \tag{86}$$

The term $\frac{\hat{u}}{\|\hat{u}\|}$ is a normalized vector with length 1. Therefore, this inner product reaches its maximum, when $\hat{u} = u^{2r+2}$. Since the inequality must hold for all $\hat{u}$, it must hold for $\hat{u} = u^{2r+2}$ as well:

$$\begin{aligned}
\max_{\hat{u}} \left(\frac{\hat{u}}{\|\hat{u}\|}, u^{2r+2} \right) &= \left(\frac{u^{2r+2}}{\|u^{2r+2}\|}, u^{2r+2} \right) = \frac{(u^{2r+2}, u^{2r+2})}{\|u^{2r+2}\|} = \frac{\|u^{2r+2}\|^2}{\|u^{2r+2}\|} \\
&= \|u^{2r+2}\| \leq \delta_q \|u^0\|.
\end{aligned} \tag{87}$$

Now we have an estimate of the error norm at the end of a Multigrid cycle compared to its norm at the beginning of the cycle.

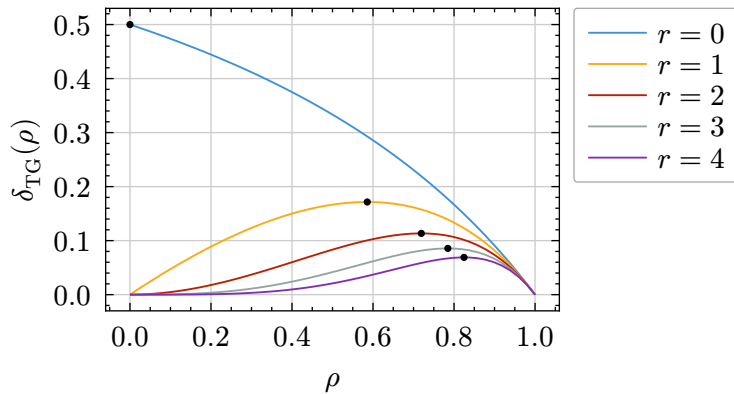
10. Convergence of the Two-Grid Method

When we only have two different levels, we observe the following: On the coarser level, we solve for the error in step 3 of the Multigrid algorithm exactly. Therefore,

$$0 = \|v_1 - u^{q-1}\| < \delta \|u^{q-1}\| \quad \Rightarrow \quad \delta = 0. \tag{88}$$

In this case,

$$\delta_{\text{TG}} := \delta_q = \max_{0 \leq \rho \leq 1} \left[\rho^r \left((1-\delta) \frac{1-\rho}{2-\rho} + \delta \right) \right] = \max_{0 \leq \rho \leq 1} \underbrace{\left[\rho^r \frac{1-\rho}{2-\rho} \right]}_{\delta_{\text{TG}}(\rho)}. \tag{89}$$



The maxima of $\rho_{\text{TG}}(\rho)$ can be found analytically by setting its derivative to zero. However, the resulting term for the maximum is rather complicated. Braess found a good upper bound:

$$\rho_{\text{TG}} < \frac{1}{(r+1)e}. \quad (90)$$

11. Convergence of the Multigrid Method

Convergence rate on level $q - 1$ must be the same as on level q when $q > 1$.

In case $\delta = \delta_{\text{MG},\mu}^\mu$, we find

$$\begin{aligned} \delta_{\text{MG},\mu} &= \max_{0 \leq \rho \leq 1} \rho^r \left(\left(1 - \delta_{\text{MG},\mu}^\mu\right) \frac{1 - \rho}{2 - \rho} + \delta_{\text{MG},\mu}^\mu \right) \\ &= \max_{0 \leq \rho \leq 1} \rho^r \left(\frac{1 - \rho}{2 - \rho} + \delta_{\text{MG},\mu}^\mu \left(1 - \frac{1 - \rho}{2 - \rho}\right) \right) \\ &= \max_{0 \leq \rho \leq 1} \rho^r \left(\frac{1 - \rho}{2 - \rho} + \frac{\delta_{\text{MG},\mu}^\mu}{2 - \rho} \right) \\ &= \max_{0 \leq \rho \leq 1} \rho^r \left(\frac{1 + \delta_{\text{MG},\mu}^\mu - \rho}{2 - \rho} \right) \end{aligned} \quad (91)$$

This equation can be solved numerically.